# Designing management education: Practice what you teach

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*In management education the medium can be the message. Students can experience the concepts they are learning if the curriculum is organized and run according to the management and organization principles being taught. This article defines ideas and presents guidelines for the design-in-the-large of education in management and organization. These ideas are then applied to two design-in-the-small cases: the XB design for an undergraduate course in organizational behavior and the thesis ring design involving collaborative supervision of thesis projects. Finally, the implications of practicing what you teach for education are discussed.*

KEYWORDS: *classroom as organization; delegation of tasks; design-in-the-large; design-in-the-small; experiential learning; interactive learning environments; learner-centered design; peer assessment; peer mentoring; thesis supervision*

Students can experience the concepts they are learning if the curriculum is organized and run according to the management and organization principles being taught. Lecturers and instructors in management subjects can practice what they teach. This study explores how experts in organization and management can present their knowledge not only as course content but also as the structure of students' learning activity. At once outside observers and inside participants, students cite and site: The organization provides both their lexicon and their job. Managerial experience and organizational scholarship give us precepts for designing and running organizations. We should design and run our learning organizations according to these precepts. This article contends that teaching of organization and management should be based on concepts of management and organization theory rather than on the traditional magisterial design of "the sage on the stage."

The theory and practice of how to create experiential and interactive learning environments have been widely studied (e.g., Burns & Gentry, 1998; García-Carbonell, Rising, Montero, & Watts, 2001; Klabbers, 2001; Kolb, Rubin, & McIntyre, 1984). Fenwick (2001) traced experiential learning through its modern roots (Garforth, 1966) to Aristotle and surveyed a broad and expanding field. Three characteristics define its modern form: involvement of the whole person (intellect, feelings, and senses),

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recognition and use of the learner's experience, and continuing reflection on experience (Andresen, Boud, & Cohen, 1999).

Neglecting this existing body of knowledge on experiential and interactive learning, scholars in organization and management by and large use traditional, instruction-centered teaching and magisterial supervision. But management and organizational behavior principles apply to educational undertakings as well as to business organizations. So, scholars in organization and management could draw on the stock of knowledge developed in their discipline by designing learning settings—by setting goals, developing goal realization processes, delegating tasks, and so forth. In other words, organization/management and education, although separate disciplines, converge in designing and organizing education in organization and management subjects.

The idea of organizing and managing education as a living experience for students began with Cohen (1993) and has been developed primarily in the context of classroom teamwork (e.g., Frost & Fukami, 1997; Griffith, 1999; Lerner, 1995). This study takes a further step in this context, exploring design ideas and propositions grounded in the living experience principle and tested in educational practice.

The argument is organized as follows: First, several design-in-the-large (DIL) principles for teaching management and organization subjects are proposed. Next, two specific applications of these principles are described: the XB design for an undergraduate course in organizational behavior and the thesis ring design for collaborative supervision. These are examples of designs-in-the-small (DISs) that draw on the more general propositions previously discussed. Finally, we discuss the implications of these design principles for management education.

## Design-in-the-large of management and organization education

This section discusses five precepts for the DIL of experience-based and interactive-learning environments in (undergraduate) education in organization and management subjects: (a) Design education as an authentic organization, (b) exploit benefits of peer mentoring and assessment, (c) give students both management and learning roles, (d) act and delegate as a senior manager, and (e) program to create constructive tension to learn.

### Design education as an authentic organization

Students, similar to all other human beings, learn certain things best through first-hand experience: "We learn to walk, ride a bicycle, drive an automobile, and play the piano by trial and error: we act, observe the consequences of our action and adjust" (Senge, 1990, p. 313). Carl Rogers (1979) argued that anything taught to someone else is relatively inconsequential and will have little influence on that person's behavior. According to Rogers, only self-discovered or self-appropriated learning significantly influences behavior (see also Garforth, 1966).

Higher education tends to use substitutes for direct, firsthand experience—cases, games, simulations, and so forth (e.g., Hassall & Joyce, 1997; Klabbers, 2001; Wolfe & Fritzsche, 1998). Thus, students experience, explore, study, and "apply" ideas, concepts, theories, and methods in the context of surrogate experiences. For students who have never worked, this structural remove can create a psychological distance from the intense reality that is the hallmark of management and organizational experience. Education as organization prevents this psychological distance from emerging because students deal with their educational environment as the organizational setting they are studying and participating in.

Designing education as an authentic organization also requires substantial task interdependency. Without it, learning reverts to a largely individual process, as in instruction-centered education. Task interdependency, students with different responsibilities working together, gives rise to organizational processes and organizational behavior, the points of reference for learning.

## Exploit benefits of peer mentoring and assessment

A well-known principle in the work of Piaget (1970) is the importance of social interactions among students as an important resource for teaching and learning. Piaget considered cooperation among peers as important as the students' interaction with teachers for intellectual development. Discussion and cooperation among students can raise awareness of different points of view, and because they communicate on the same cognitive level, peers may have more influence than a teacher in helping a student abandon deeply ingrained misconceptions and convictions (Cole, John-Steiner, Scribner, & Souberman, 1978; Ginsburg & Opper, 1969; Mouton & Blake, 1984; Piaget, 1970; Strom, Strom, & Moore, 1999).

Reciprocal teaching and peer mentoring produce significant learning benefits compared to teaching and mentoring by people of higher hierarchical status (e.g., Bruffee, 1993; McDougall & Beattie, 1997). For example, students who often engage in peer group discussions about their work in mathematics and science courses perform significantly better than students who study alone out of class (Bruffee, 1993; Davidson, 1990).

The second idea therefore is to reinforce and exploit social interactions among students, particularly through peer mentoring and assessment. Having peers teach and assess one another creates many opportunities for learning from experience how to observe and understand organizational behavior, communicate effectively, resolve conflict, motivate, assess, give feedback, and lead.

## Give students both management and learning roles

Most undergraduate majors seek to train their students to think critically, and to consider any issue from two or more perspectives before drawing conclusions (Association of American Colleges, 1990). Management educators strive for similar nimbleness explicitly through concepts of framing (Bolman & Deal, 1997) and different

images of organization (Morgan, 1996) and implicitly through the very existence of different functions of management.

Taking different roles may reinforce cognitive differentiation and help "declassrooming the classroom" (García-Carbonell et al., 2001, p. 485). By taking both management and learning roles, students directly experience the organizational principles of specialization and coordination. Their peers in various roles make demands on them. They notice administrative processes and understand the role-based necessity of exercising authority because their peers—real people in contrast to the alien teacher creatures (so to speak)—do it. In other words, by taking both management and learning roles, students have to relocate the locus of conversational control from the teacher to themselves (cf. Scarcella & Crookall, 1990).

## Act and delegate as a senior manager

Modern organizations delegate important tasks to the lowest practical level, as seen in the practice of 360° evaluation (Tornow & London, 1998). In turn, initiatives by subordinates and emergent events challenge managers in any organization. Delegation of key tasks allows such initiatives and events to happen in the classroom. Management educators, just like real-world managers, can learn from the challenges of delegating real responsibility, dealing with departmental conflict, facing the anxiety arising from assessment processes and outcomes, and so forth.

In light of the preceding ideas, this principle suggests that the teacher should take on the role of senior manager of the organization and delegate everything he or she possibly can. Students should participate in all key activities, including supervision and assessment. With regard to educational outcomes, the senior manager remains responsible to superiors and other stakeholders, just as the chief executive officer remains accountable to the board of directors and other (external) stakeholders.

Delegation does not imply doing almost nothing; delegation of numerous tasks gives rise to a new task domain: coaching and advising students. Thus, as in many real-world organizations, the senior manager will have to spend time with students in relatively informal settings to help them develop learning objectives or research questions, research and action plans, or effective ways to teach ideas and concepts to others. Acting as a manager may bring immediacy to the topic. As experts, senior managers can recognize and point out managerial and organizational issues confronting the organization. Theory comes alive.

## Program to create constructive tension to learn

In educational settings, delegating without well-defined responsibilities and tools may lead nowhere. Thus, the final design idea, simple and straightforward, is: Program! During the late 1960s and early 1970s and still occasionally today, teachers ask students to design their own classroom organization and then experience its consequences. Learning-centered education is often confused with relatively high degrees of freedom and discretion of learners (e.g., Rogers, 1979) and called "unstructured."

Of course, in all learning-centered education students have more responsibility and discretion than in instruction-centered education. But delegation and structure are entirely separate ideas. And most management texts tell managers to articulate job assignments clearly.

Management educators should design tools and procedures to create challenging but achievable learning for students delegated responsibility. If students perceive an unbridgeable chasm between their competences and attitudes at the beginning and those expected by the end, this unmanageable gap will confuse and frustrate them (Burns & Gentry, 1998). Students cannot be expected to reinvent every wheel in a complicated machine.

## Overview

Taken together, these five design precepts at the level of DIL intertwine basic concepts of the operational system with those of the management system. Thus, learning organizational behavior and teaching (or managing) organizational behavior unite in education as organization. The sociotechnical systems approach has long advocated such interaction and synergy between the operational and management system (e.g., Emery, 1989; Miller & Rice, 1967). Sociotechnical systems theory however has failed to capture the dynamic, unpredictable interaction between organizations and their external environments (Pasmore, 1995). Our design precepts build into both operational and managerial systems change arising from internal, self-referential (rather than external) sources.

In this respect, the self-referential correspondence between DIL and DIS involves the correspondence among goals and processes at both levels. At the level of DIS, the *first-order goal* of education as organization is to learn concepts of organizational behavior and management and assess learning outcomes. For example, in the XB case (discussed more extensively in the next section) students use new vocabulary to describe the situation they face: They refer to the instructor as a senior manager whose written instructions (formalization) enable him or her to delegate or decentralize authority using a laissez-faire management style (standing aside and letting events happen—as in all simulations). But the organization's high horizontal differentiation—having a variety of functions at the same level—makes for complexity and creates an authority vacuum. Each participant must accomplish an objective through the work of others (per the definition of management). Some may try using the rules of the formal organization and grading power to get others to do assigned tasks, to control, and to evaluate. Others rely on informal social links and trust their peers to do their jobs. Neither approach works with everyone. As norms (unwritten rules) in the organization's culture change, participants learn that by giving and receiving frank feedback (not taking things personally) they can use influence (personal power) to motivate. Responding to the inadequacy of formal rules in their environment, participants exercise leadership. Whereas in traditional management courses students typically define words on tests ("Motivation is . . ."), in education as organization they use course vocabulary to describe and change processes around and within them.

At the level of DIL, the *second-order goal* of education as organization is to develop an authentic organizational context for these learning processes that participants manage and change. Integrating the operational and management systems usually makes the distinction between students and instructors disappear (almost) completely: Both student and instructor are managing, supervising, learning, assessing, and so forth. External accountability for learning outcomes and their measurements is—in theory—the only domain that the instructor operates in from an allopoetic (external) position (cf. Klabbers, 1999).

From a process perspective, the DIL precepts described earlier focus on the mental model of the educators to help them prepare and implement any specific DIS in this area. This mental model is rather general in nature and is relevant for (at least) all those educational settings in which learning and teaching intertwine—as operational and managerial activities. The DIL as a mental model precedes the implementation of DIS practices in the sense that the educator first needs to become familiar with this unconventional perspective on education.

Any specific DIS in this area will serve as a tool to create actual systems with real people, behavior, learning outcomes, and so forth. Such actual systems are also contingent on a specific educational context (e.g., an elective course in an undergraduate program or thesis supervision in a master's program). The two examples at the level of DIS discussed in the next section examine whether the self-referential quality of education as organization can be realized in practice.

## Design-in-the-small: Two cases

This section will describe the characteristics and results of two DIS designs that serve to illustrate how the general ideas developed in the previous section can be elaborated and applied in coherent practice. These two designs are the XB design for undergraduate organizational behavior courses and the thesis ring for supervising thesis projects. Developed at Saint Michael's College (Vermont, USA), the XB design has been adopted by at least eight other universities and colleges in the United States, Australia, and Europe. Thesis rings were developed at Maastricht University (the Netherlands) and subsequently adopted by five other European universities. Both designs therefore appear to be transferable.

At the level of DIS, specific choices have to be made regarding the conditions created for the participants. The conditions created in the XB case are similar to those of a functional hierarchy with lots of teamwork. The conditions created in the thesis ring are similar to those in the academic community, involving peer review and writing papers.

### The XB design

XB (The eXperience Base) was developed by Roger Putzel at Saint Michael's College for an undergraduate course in management and organizational behavior. The XB

design brings an immediate experience of organizational behavior to students—in areas such as organizational design, communication, leadership, conflict resolution, and giving and receiving feedback—by treating the classroom as an organization. Putzel (2003) described his mission in developing this course as

> experience-based learning using the concepts and tools of management and organizational behavior. XB is a laboratory, a place for open-ended experimentation (not merely for repeating previously conducted experiments). Management is not a subject to learn and half forget, like other subjects we study. It is a set of principles, skills, beliefs, attitudes, and values that can enrich our lives and help us accomplish more than we ever dreamed possible, working together in organizations. (p. 54)

XB's process-oriented or self-referential approach allows participants to convert real events into learning opportunities. XB is a complex organization whose product is learning. Participants learn the concepts, skills, and attitudes of a perspicacious and effective manager. More specifically, the XB organization aims to improve its members' ability to (a) list and define concepts, (b) articulate thoughts, (c) use intellectual tools, (d) recognize social (external) situations, (e) recognize personal situations, (f) act on their own initiative using resources at hand to achieve their own objectives, and (f) make first-time mistakes (Putzel, 2003). The XB manual describes the last product, first time-mistakes, as follows: "People in successful XB divisions (sections) make a lot of them; in less successful divisions they don't dare to make so many" (Putzel, 2003, p. 4). The XB organization values first-time mistakes, but repeated mistakes indicate slow or no learning. In short, the XB design seeks to create a liberating structure (Torbert, 1978) that reduces participants' fear of failure yet rewards their learning initiatives.

At the beginning of the course, the XB organization exists only on paper, in a manual of 280 pages describing goals, objectives, structure, rules, job descriptions, and so forth (Putzel, 2003). This manual also contains a large number of detailed references to classic as well as more recent organization and management theories, ranging from Anthony (1988) and Argyris (1957) to Weisbord (1988) and Yates (1989). The organizational structure with four departments and 12 groups is reproduced in Figure 1. The departmental structure of Responsibility, Doing, Observing and Understanding stems from Kolb's learning cycle (Kolb et al., 1984). For each group (i.e., subject), the manual provides extensive descriptions of key contributions to the literature, detailed responsibilities, and intended learning outcomes that apply this subject to the immediate environment of the class organization. It directs one group, for instance, to apply the concept of trust from Douglas McGregor's works by pointing out trusting or distrusting behavior that occurs in class and by running the Prisoners' Dilemma exercise. Each group has tasks and responsibilities distinct from but interacting with those of other groups. This structure places people in CAT roles (Coach, Administrator, Teacher) to deliver learning.

In the first week of the course, the staffing group places other participants in groups and departments with their distinctive job descriptions and objectives. Immediately, students take over the organization. The teacher takes on the senior manager role,

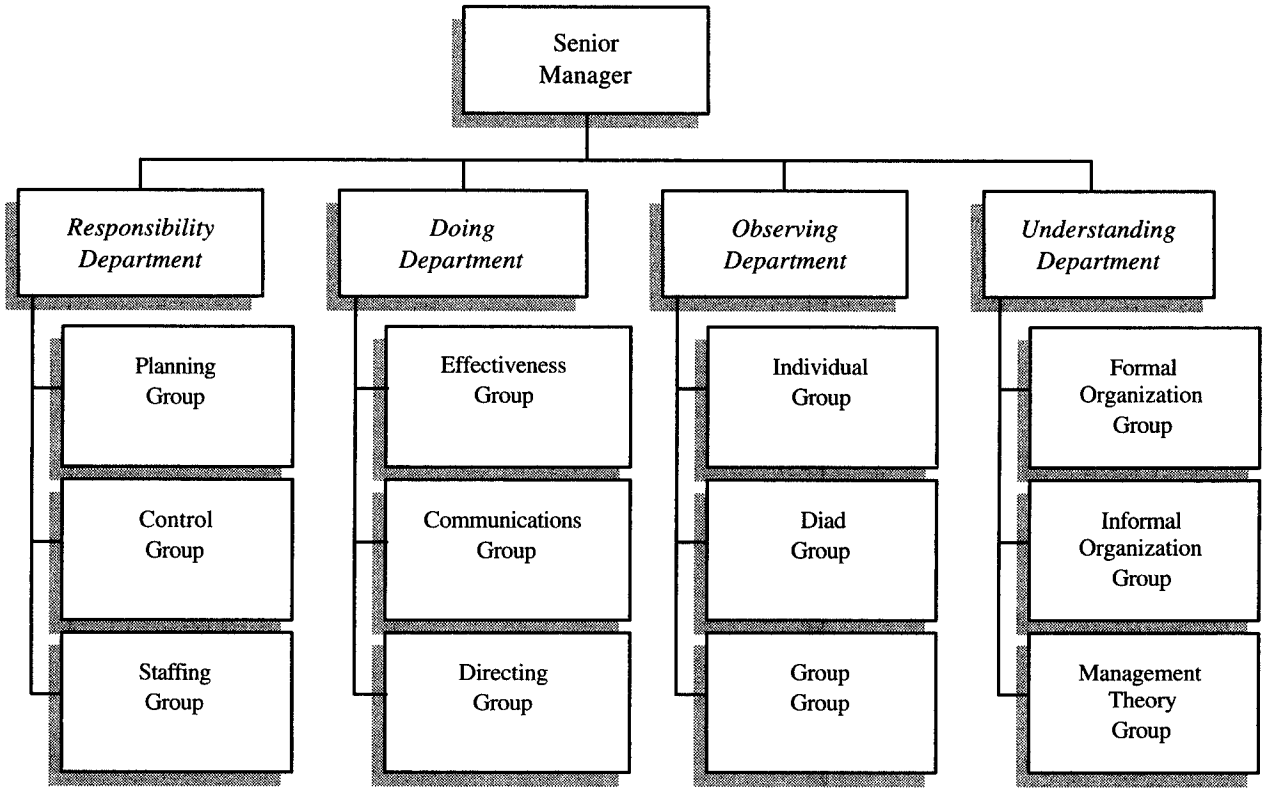


**FIGURE 1: XB's Organization Chart**
SOURCE: Putzel (2003).

delegating every possible function and activity to a department or group. Some senior managers (lecturers) even delegate grading. The senior manager coaches, often giving advice outside class, trying to stay out of the way and let students manage their functions in the organization (Putzel, 2001).

The manual and the senior manager help participants apply theories and tools to the process of developing and managing the XB organization. Each group reads its own part of the manual and other text assignments on certain topics so by the third week (in theory) the organization has done all the reading for the semester. XB must then start exploiting its internal resources. Each group has the objective of teaching members of other groups the specific concepts and skills that it has learned. The groups may reach their objectives in any way they choose: making presentations, creating ad hoc teams, approaching individuals outside class and getting them to do something differently, developing self-test questionnaires, short simulations, and so forth.

Those to whom learning is delivered by people in CAT roles are participants themselves. They wear another hat: the ALE role (Apprentice, Learner, Employee). A participant in the CAT role teaches, coaches, and exercises authority over another participant in the ALE role.

Although they can read in the manual what to do, it usually takes participants three weeks (of a semester) to get used to being held accountable for results but not being told what to do—to feel like managers rather than students. Many participants begin this course with no great sense of responsibility. They struggle to find out how the organization functions, and they learn that they—or no one—will make (their part of) XB work.

Blindly following the manual does not make this (or any other) organization work. An informal organization emerges: Some participants sit back and let others work, and then important parts of the organization do not function as they should. Groups may also fail to accomplish anything because they only know their own material: The planners do not motivate, the formal organizers do not communicate, and so forth. This disorder creates opportunities for learning, and theories and concepts of management and organizational behavior in the manual offer diagnostic tools for understanding what is going on and what to do to make the organization more effective. Groups start to recognize situations to which their skills or theories relate. When a group provides a missing concept or teaches others a relevant skill, the organization starts to work. XB's members start to learn about setting and reaching objectives, decision making, group dynamics, effective delegation, how to recover from failure, listening behavior, effective feedback, and above all, continual learning from experience. They are learning from each other in the context of their own work.

The design of the grading system empowers the participants to measure and motivate each other. It has three distinctive features: a large number of measurements, delegation, and rank ordering (Putzel, 1992). About 130 different measurements per student are collected during the semester; these copious data tend to reduce individual errors measuring a person's performance and learning. Each person receives ample feedback on the basis of detailed criteria regarding participation, reading, effort, written memos, and conceptual understanding; moreover, no single grade counts more than 2% in the final evaluation. The grading system, "like the rest of the organization, is intended to be globally robust, rather than meticulously accurate" (Putzel, 1992, p. 209). The senior manager delegates almost all grading to the participants. That is, XB members take responsibility for control (the management term for assessment) as well as learning. At the very end of the course, the senior manager assigns the final grades on the basis of data provided by the Control Group. XB members may change the assessment system but must first demonstrate that they understand and can apply the control strategy (assessment approach) described in the manual.

In most organizations, strong norms militate against frank evaluation of peers. Most people—students not the least—indulge others' shortcomings and give all peers high grades. Fearing their peers, they shirk responsibility (Peiperl, 1999). XB requires participants to grade by rank order. This at first draconian measurement system sometimes evokes huge resistance, of course, despite evidence of its effectiveness (Putzel, 2001). Once students start working with the rank ordering system, this resistance disappears; this happens especially when students outside the class who have already taken the course tell those in the course "You'll get used to it" and thereby help break dominant norms in the first sessions. In the organization, rank ordering guarantees useful distinctions for evaluation purposes. Rank ordering also produces learning of great value. When participants fudge the system, they see the organization quickly go out of control and must confront ethical issues (anticipated in the manual of course). When participants honor the system and dare to evaluate each other frankly, they discover the efficiency and benefits of social capital and grow considerably in self-confidence.

Reflections by former participants illustrate the kind of experiential learning that occurs in XB. A former student at Saint Michael's College reported:

> This course has had such an incredible impact on me, how could I possibly put it all to words? I would walk away from each class and suddenly view each circumstance that I was faced with in a completely different light. Real life instances of concepts were everywhere I turned. (Putzel, 2001, p. 3)

An education major at the same college wrote:

> This experience has exemplified to me the endless possibilities that exist to learn and to teach.... I want to be able to reach people in a new way, and Roger [senior manager in this course] has proved to me that this is entirely possible. (Putzel, 2001, p. 3)

Other universities have had similar experiences with the XB approach (Romme, 2003). Of course, students also criticize the grading system. Moreover, it sometimes takes the first groups at new sites a long time to accept the system. Friendship, culture, disorganization, and cheating can distort its results. Crises, failures, and conflicts affect not only student participants but also the teacher as the organization's senior manager, who must be willing to deal with what is really happening and learn from it.

In the very pathology of this self-referential learning environment, XB recapitulates the confusing complexity and fickle humanity of the real world. By taking over and running a complex, imperfect organization; by observing emergent events and relationships; and by learning from their successes and failures, all participants in XB experience the management and organizational behavior they are studying.

## The thesis ring design

In 1996, Maastricht University started to experiment with new ways of supervising students working on master's thesis projects. These experiments tested the use of peer mentoring and coassessment in a collaborative system managed by the senior supervisor, a ring. At this university and many other postsecondary schools and universities with large numbers of students, teaching staff spend considerable time supervising and assessing thesis projects. Through sharing supervision and assessment, the experimenters hoped to increase both learning and efficiency for students and supervisors.

Two key steps in the experiment were the adoption of rules for organizing a thesis ring and the development of an assessment procedure that would involve academic supervisors and cosupervising students. In the first year of the first thesis ring, the basic design was further developed by procedures and guidelines for decision making, chairing meetings, giving feedback, and assessing draft and final thesis reports. Several straightforward decision-making rules and procedures were adapted from circular organizing, a relatively new way of organizing and managing participation (Ackoff, 1994; Romme, 1998, 1999).

Subsequently, other professors and lecturers at Maastricht University adopted this design for supervising their (master's) thesis students. The university now operates

four permanent thesis rings in information management, organizational behavior, systems theory, and strategic management. Thesis rings in education, information management, architecture, and organization theory have also been set up at five other European universities. The thesis ring design assigns multiple roles—including the role of coach (cosupervisor), chair, scribe, and learner—to students. This design involves the following arrangements:

- The thesis ring extends to students the formal authority of at least one academic supervisor to collaboratively supervise and grade thesis projects. That is, at least one senior lecturer or professor is a permanent member.
- The domain of the thesis ring, established by the academic supervisor(s), encompasses topics supervised in the ring (e.g., organizational behavior or information management). Students can enter the thesis ring at any time during the academic year. The academic supervisor has the authority to accept new members based on considerations of capacity and domain.
- Membership ends with a candidate's successful completion of the thesis unless decided otherwise.
- Each meeting proceeds according to a standard format involving four parts: (a) an informal opening round, (b) determination of the agenda on the basis of a proposal sent out prior to the meeting, (c) discussion and feedback on progress reports regarding thesis projects, and (d) an evaluation discussion.
- A number of policy decisions are made using the consent principle. A decision has been taken by consent when no participating ring member wants to argue against the proposed decision (no argued objection). Some policy issues include work procedures (e.g., preparation of agenda), election of the chairperson and scribe, and thesis assessment. External authorities (i.e., exam committees in European universities) typically establish the assessment standards and criteria. Use of the consent rule gives power to each participant in the ring regardless of status.
- Both the chairperson and scribe are chosen for a limited period (e.g., 3 months) by an election procedure based on the consent rule.
- Students usually act as chairperson and scribe. Only in exceptional cases will academic supervisors chair.
- The final assessment of each thesis is done in a meeting of the thesis ring in which at least two staff members participate, including the permanent academic supervisor and an external academic assessor. The external assessor, who must not have been involved in the preceding supervision process, can therefore focus entirely on the end product—the final thesis—without being biased by the production process.
- The procedure for assessing the final thesis is based on the following ideas. First, only those (co)supervisors who prepare the assessment adequately—using a standardized assessment form—participate in the assessment. The ring's secretary makes sure they have all information required to do so. Second, the assessment takes place on the basis of arguments rather than authority. Third, to achieve openness, the assessment procedure starts by minimizing the interaction between the initial individual evaluations to acknowledge any differences in opinion and judgment. Subsequently, stepwise argumentation and dialogue bring the ring to a final grade that is both well argued and broadly accepted.

Detailed procedures and guidelines for decision making, chairing meetings, giving feedback, and assessing thesis reports have been published in a 60-page manual (Romme & Nijhuis, 2000; see also www.personeel.unimaas.nl/j.nijhuis/kringen/index.htm).

The collaborative nature of the thesis ring design implies that each student has two distinct roles: as a thesis writer being supervised, advised, and assessed and as a cosupervisor, coadvisor, and coassessor of others. As a result, students switch roles at least once every meeting. An empirical study of six thesis rings (Rompa & Romme, 2001) showed that the traditional distinction between the roles of supervisors and students largely disappears in these rings. In conventional interaction, the authorized expert supervisor leads an individual student through the research and writing process. In thesis rings, collaborative supervision and learning by students and senior manager(s) replaces the supervisor's monopoly on feedback and grading. Now, students rather than staff chair and moderate discussions, coordinate information flow between participants, and organize meetings (Rompa & Romme, 2001).

A student at Maastricht University described the most important advantage of thesis rings compared with the usual one-to-one supervision:

> In the beginning I was impressed by the extra work I had to do, reading drafts of other students, and the like. Later, however, I began to understand that I only had to write one thesis but was learning as if I were writing seven theses. (Rompa & Romme, p. 307; translated by authors)

An academic supervisor at Maastricht University added detail:

> A student in my thesis ring learns from the mistakes and the successes of fellow students; the overall attitude of students with regard to the quality of their own written work as well as the work of others changes to a much more critical attitude, because of the fact that one also reads and discusses the work of others. (Rompa & Romme, p. 307; translated by authors)

The thesis ring design also builds supervisors' awareness of the implications of their actions, particularly the effectiveness of their interventions in students' learning. Four supervisors commented as follows on their experiences in their own thesis ring:

> "When, as a supervisor, you start talking nonsense, or when the thesis writer does not understand what you're saying, the chair or one of the other students will correct and help you."
>
> "As supervisor, I get a more complete picture of the qualities of the students I'm supervising, because I also see them in interaction with others."
>
> "Even when a student does not submit a written text, he is still involved as a result of his supervision task. The criterion for interaction is thus not the submission of a piece of work, as is usually the case in traditional supervision of thesis students."
>
> "The quality of a thesis is viewed from multiple viewpoints. As a result, more problems and details are picked up than when only one or two people do the supervision" (Rompa & Romme, pp. 307-308; translated by authors).

A comparative analysis of questionnaire data regarding interaction behavior in six thesis rings, in several other cooperative methods to supervise thesis projects, and in a large number of traditional (individual) supervision cases led to the following findings: Both effectiveness-increasing interaction behavior (e.g., constructive

argumentation and comments expressing confidence and trust) and effectiveness-decreasing interactions (e.g., hostile comments and interruptions when someone else is speaking) were found to be much more prevalent in thesis rings than in the two other supervision modes (Rompa & Romme, 2001). Despite the more ambiguous and perhaps complex nature of supervision in thesis rings, there were no significant differences between the three supervision modes in terms of students' satisfaction (Rompa & Romme, 2001). It is therefore possible without sacrificing the quality of supervision to include students in supervising, advising, and assessing their peers.

## Implications for management education

This section will evaluate the XB and thesis ring cases in the context of the design propositions for management and organization education outlined earlier: Design education as an authentic organization, exploit benefits of peer mentoring and assessment, give students both management and learning roles, act and delegate as a senior manager, and program to create constructive tension to learn.

### Education as organization

Dealing with education as an authentic organization apparently requires a shift in the learning orientation of both educators and students from what Klabbers (2000) defined as "learning as acquisition" to "learning as interaction." This fundamental shift forces learners to deal with the dual structure of their (educational) systems, that is, to "learn to produce and self-produce structures as a context for action" (Klabbers, 2000, p. 380).

In the context of this epistemological shift, the two DIS cases explored in the previous section appear to create intense experiences. Beyond simulating organizations, as with computers or role-plays, the first principle implies setting up a real organization that produces learning behavior leading to the intended learning outcomes or products. From a simulation perspective, the cases we have discussed involve a real-time simulation of a self-referential process of organizing and managing students' learning. That is, the organization is only real as far as its broader educational context is concerned; outside the educational setting, education as organization cannot exist. In this respect, the organization has an educational goal, to produce learning, and uses managerial and organizational tools, procedures, and systems to reach its goal.

Designing education as organization requires substantial task interdependencies between participants. Without task interdependence and shared responsibility, learning may revert to a largely individual process, as in instruction-centered education. Task interdependence and shared responsibilities give rise to organizational processes, behavior, and learning.

The DIL precepts to design education as an authentic organization apparently apply to all educational settings in the area of organization and management subjects; in this area, the first-order goal of the operational system (e.g., learning organizational

behavior concepts) and the second-order goal of the managerial system (e.g., managing and changing the organization) become self-referential. Some subjects outside this domain—for example, history or physics—are not likely to benefit from these design precepts because the self-referential quality between managerial and operational system cannot be realized. For other subjects, this self-referential quality can perhaps be built into the educational system—for example, education as contract for courses in contractual law or education as cognition for learning and teaching cognitive psychology. In any case, the DIL precepts we have described may then have to be adapted.

At the level of DIS, the XB design cannot be transferred to educational settings other than organization and management subjects. The thesis ring design however has been effectively applied to subject areas other than management/organization, such as architecture, market research, and educational theory (at the University of Delft, Maastricht University, and University of Leuven). Future research needs to address the question of whether and how thesis rings in these settings suffer from a lower synergy between first- and second-order goals of the thesis ring as an educational system.

## Peer mentoring and assessment

Peer mentoring and assessment can be considered as uses of a resource in an organization. Successful organizations appear to exploit their unique resources more effectively than other organizations (Combs & Ketchen, 1999). At many universities, academic staff members perceive undergraduates as distractions from research, neglecting their potential as resources for teaching. The educational designs discussed in the previous section require that we acknowledge and involve students as potential supervisors, team leaders, coteachers, coassessors, and so forth. Compared to other disciplines, faculty in organization and management studies may have a unique advantage: Their discipline mandates the efficient use of human resources and advocates empowerment at the base of the organization. They need only practice what they teach.

## Students have both management and learning tasks/roles

In both XBs and thesis rings, taking different roles tends to reinforce cognitive differentiation. For instance, when a student chairing a meeting asks the senior manager/supervisor to let the previous speaker complete her thought, another student may distinguish for the first time between age and authority. The thesis ring and XB designs get students to consider issues in more than one way by having them play different roles at different times. Both designs put participants in both supervisory/administrative roles and learning/content roles.

Perhaps more important, participants in XBs or thesis rings learn to make the distinction between being a person and filling a role, a distinction that will serve them well in their careers. In their roles, they assess and grade their peers and learn not to take criticism (too) personally. Conversely, they see that a role by itself does not confer adequate power, that personal power must supplement legitimate power. Moreover, taking management as well as learning roles appears to encourage students to manage

their own learning, a fundamental value of experience-based adult learning (Knowles, 1975; Taylor, 1998).

## Acting and delegating as senior manager

The senior manager role for teachers presents its own challenges, costs, and risks, notably, accountability, time demands, the unknown, and the challenge to emerge as a person from behind a role. The senior manager remains responsible and accountable to superiors and other stakeholders for educational outcomes (see earlier). When a student forgets to plan a crucial meeting, lets a discussion wander out of control, or serves alcohol illegally (it happened), the senior manager must answer for it. A more quotidian concern arises in that organizations sometimes work slowly and novices' many mistakes take time to rectify. In practice, we have found that senior managers in an XB organization or thesis ring, similar to any other manager, often regret having delegated!

Moreover, coaching as a senior manager, although gratifying, tends to take more time than teaching. Worse yet, having delegated so much—and despite the distractions of coaching—the senior manager will hear industrial history knocking on the door. What value does the senior manager add? With teaching functions delegated, what remains of our job? The senior manager apparently faces the unknown, the unprogrammed side of being a manager, perhaps its most daunting one. In this respect, working with students in organized settings such as XB apparently requires openness and vulnerability. Compared to the traditional lecturer, the senior manager reveals more of his or her personality. A similar process is taking place in many business organizations, where according to Hirschhorn (1997), "one cannot hide behind the role of leader; instead, one must bring more of one's passions, fears and values into it" (p. 21).

Acting and delegating as a senior manager have advantages that offset these challenges, starting with the fundamental advantage of managing over doing everything yourself: When the organization works, it accomplishes more than an individual and takes many burdens off the senior manager. The senior manager does not have to attend all meetings, which also run—students often say "better"—without the senior manager. One XB class ran productively unsupervised for three weeks when the senior manager faced a family crisis. Thesis ring meetings held without senior supervisors are said (by students) to be the "best ever." In many XB and thesis ring situations, students manage so visibly that the senior manager must occasionally remind them of his or her own managerial role and dilemmas lest they not learn what successful delegation looks like from above.

## Program to create constructive tension to learn

Well-developed procedures and responsibilities to create a constructive tension between initial and expected competences and attitudes appear to be important. The XB and thesis ring designs suggest that the organizational structure must serve as a vehicle for learning. In management terms, XB and thesis rings provide a great deal

of structure because organization theory tells us that formalization makes decentralization possible (e.g., Adler & Borys, 1996). The programmed responsibilities and procedures do create limited and constructive tension between their competences and attitudes at the beginning and those expected by the end.

The many procedures in these cases seem to clash both with the antibureaucratic trend of recent management literature (cf. Senge, 1990) and with open learning theories in the educational literature (e.g., Paine, 1988; Rogers, 1979). However, the practice of XBs and thesis rings suggests that formal responsibilities and procedures enable collaboration and participation. Programmed procedures help newcomers get started in the organization; moreover, when participants have become familiar with how the (XB or thesis ring) organization actually works, they start redefining and adapting procedures and responsibilities.

Procedures and responsibilities also produce constructive interpersonal tension by limiting the scope—though not necessarily the intensity—of interpersonal conflict. In open, unstructured learning students delve into personal problems, but they rarely resolve them in a college course. Conflict, which happens in most organizations, may become excessively personal if there is nothing else to fight about. In a thesis ring, students discuss and debate the content and conduct of research. In XB, they get very angry about missed meetings and unjust ranks (that determine less than 2% of their course grade). They experience the negative emotions of real-world organizations, but these are tempests in a teapot. Formal responsibilities and procedures are the teapot.

Programming responsibilities and procedures for authentic organizational designs may require a huge investment of time and other resources, which may discourage many lecturers and professors. Once made however, such an investment enables lecturers to assign completely new tasks and roles to students without overwhelming and confusing them. Moreover, if the pioneers make tested, coherent designs available in books and manuals and on web sites, others can adopt and then help elaborate them.

## Conclusion

Education and management, although distinct disciplines, converge in the design of education in organization and management. Curiously, scholars in organization and management have largely neglected the precepts of their own discipline by using traditional, instruction-centered teaching and magisterial supervision. Management and organization principles apply to educational as well as to other organized settings.

In this article we first described several propositions for designing (in-the-large) education in management and organization. Subsequently, these ideas were applied to and elaborated for two DIS cases: the XB design for an undergraduate course in management and organizational behavior and the thesis ring design involving collaborative supervision of thesis projects. Finally, the implications of practicing what you teach for management education have been discussed. The five design propositions developed and illustrated in this study are the following:

- design education as an authentic organization,
- exploit benefits of peer mentoring and assessment,
- give students both management and learning roles,
- act and delegate as a senior manager, and
- program procedures and responsibilities to create constructive tension to learn.

These design principles correspond with literature on learner-centered education as well as the literature on management and organization design and have been tested in two experiments described in this article. The two cases, XB and thesis ring, are two among many possible designs that can be produced with this set of design precepts. At present, education as organization precepts are most easily applied to educational settings involving management and organization(al behavior) subjects. The self-referential nature of education as organization applies more easily for these subjects than for others. Moreover, each application at the level of DIS will produce its own distinct processes and outcomes. XB replicates a traditional organizational setting, whereas processes in thesis rings resemble academic writing, review, and debate. The processing of meaning of concepts—such as decision making, leadership, personality, motivation, and power—will vary in these different settings.

Future work might develop and test education as organization designs in other educational contexts. Moreover, going beyond innovations implemented by individual scholars, the DIL propositions described in this article can also be used to shape a complete program in business or management studies.

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